

Retry and Circuit Breaker

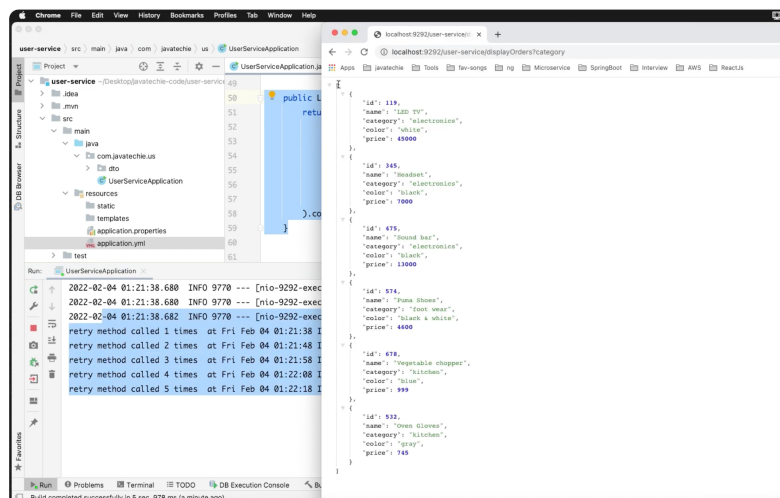
<https://github.com/Java-Techie-jt/springboot-resilience4j/tree/main>

```
13
14     resilience4j:
15     circuitbreaker:
16         instances:
17             userService:
18                 registerHealthIndicator: true
19                 eventConsumerBufferSize: 10
20                 failureRateThreshold: 50
21                 minimumNumberOfCalls: 5
22                 automaticTransitionFromOpenToHalfOpenEnabled: true
23                 waitDurationInOpenState: 5s
24                 permittedNumberOfCallsInHalfOpenState: 3
25                 slidingWindowSize: 10
26                 slidingWindowType: COUNT_BASED
27
28
29     retry:
30     instances:
31         userService:
32             maxRetryAttempts: 5
33             waitDuration: 10s
```

Refer line number 41 and 42.

```
39
40 @GetMapping("/displayOrders")
41 // @CircuitBreaker(name = USER_SERVICE, fallbackMethod = "getAllAvailableProducts")
42 @Retry(name = USER_SERVICE, fallbackMethod = "getAllAvailableProducts")
43 public List<OrderDTO> displayOrders(@RequestParam("category") String category) {
44     String url = category == null ? BASEURL : BASEURL + "/" + category;
45     System.out.println("retry method called " + attempt++ + " times " + new Date());
46     return restTemplate.getForObject(url, ArrayList.class);
47 }
48
49
50 public List<OrderDTO> getAllAvailableProducts(Exception e){
51     return Stream.of(
52         new OrderDTO(119, "LED TV", "electronics", "white", 45000),
53         new OrderDTO(345, "Headset", "electronics", "black", 7000),
54         new OrderDTO(475, "Sound bar", "electronics", "black", 13000),
55         new OrderDTO(574, "Puma Shoes", "foot wear", "black & white", 4600),
56         new OrderDTO(678, "Vegetable chopper", "kitchen", "blue", 999),
57         new OrderDTO(532, "Oven Gloves", "kitchen", "gray", 745)
58     ).collect(Collectors.toList());
59 }
60
61
62 public static void main(String[] args) {
63     SpringApplication.run(UserServiceApplication.class, args);
64 }
65
66 @Bean
67 public RestTemplate restTemplate() {
68     return new RestTemplate();
69 }
70 }
```

After 5 times we got default (fallback) response. Notice retry number in the log.



When downstream is up after 4th retry – we got original response.

